SALASEL ENTERPRISE

SYSTEM BLUEPRINT & ARCHITECTURE SPECIFICATION

Industry Target: Supply Chain / Retail Tech / B2B E-commerce
Core Technology Stack: Flutter (Mobile), Angular (Web Dashboard), .NET Core (Backend API), SQL Server & pgvector
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1. Enterprise Design System & Color Matrix

To guarantee exact visual consistency across Mobile and Web applications, all interface screens, layout feedback components, and transaction states must strictly adhere to the following color token map:

Color Token	Hex Code	Functional Application Context
Primary Blue / Orange	#0052CC / #FF5630	Trustworthy B2B primary interactions, high-intent action buttons, key setup routes
Success Green	#36B37E	Confirmed transactional workflows, valid database commitments, complete catalo
Warning Yellow	#FFAB00	Low asset stock level notification badges, processing loading state shimmers, pe
Critical Error Red	#DE350B / #FF5630	Fraud flag exceptions, system connection failures, network constraints, missing h

Typography: The platform enforces native Right-to-Left (RTL) Arabic text rendering using specialized optimizations like **Cairo**, **Tajawal**, or **Almarai**. This must flip instantly to left-aligned formatting when users switch to the English (LTR) mode using **Inter**.

2. Fully Comprehensive Platform Sitemap

```
                              SALASEL ENTERPRISE ARCHITECTURE MAP
                                              |
         +----------------------------------------------+-------------------------------
-----+
         v                                    v
     v
[Public Web Ecosystem]             [Merchant Mobile Application]          [
Supplier Web Control Tower]
 (Angular UI Framework)                  (Flutter Cross-Platform)             (
Angular Enterprise Client)
         |                                    |
     |
         |-- 1.0 Marketing Showcase            |-- 1.0 Enterprise Onboarding
     |-- 1.0 Gateway Auth Portal
         |-- 2.0 Architectural Trust          |-- 2.0 Core Workspace Hub
     |-- 2.0 Live Central Feed
         |-- 3.0 Public Supplier Directory     |-- 3.0 Voice Procurement Loop
     |-- 3.0 Kanban Order Operations
         |-- 4.0 Open Integration Center       |-- 4.0 Local Inventory Ledger
     |-- 4.0 RAG Knowledge Base Center
         +-- 5.0 Global Contact & Lead Portal  |-- 5.0 Split-Fulfillment Engine
     |-- 5.0 Fraud & Hard-Limits Panel
                                              +-- 6.0 System Resilience Guard
     |-- 6.0 Internal System Audit Room

     +-- 7.0 Business Intelligence Hub
```

2.1 Public Web Ecosystem (Angular - Client-Facing)

1.0 Marketing Showcase Hub:

* 1.1 Hero Banner (Embedded with interactive Egyptian Arabic voice ordering simulation dashboard and fluid waveform visualizer).
* 1.2 Core Friction Mapping (Visualizing manual WhatsApp gridlock/chaotic calls vs. Salasel structured data conversion).
* 1.3 Ecosystem Breakdown (FMCG retail flow vs. compliant batch-tracked pharmaceutical routing structures).
* 1.4 Enterprise Metric Ticker (Live counts of total orders routed, active merchants, and system-wide catalog SKUs processed).

2.0 Architectural & Strategy Trust Portal:

* 2.1 The AI Trinity Explainer (Deep-dive on Amazon Transcribe + Claude Haiku/Sonnet + LangGraph autonomous states).
* 2.2 High Availability Engineering (Diagramming the ITI API Access Gateway paired with the automated Groq API fallback framework).
* 2.3 Data Isolation Policy (Technical brief explaining Row-Level Security (RLS) configurations and database isolation).

3.0 Public Wholesaler & Supplier Directory:

* 3.1 Indexed Directory of Registered Distributors featuring dynamic performance and fulfillment reliability ratings.
* 3.2 Supplier Onboarding Lead Form (Capturing enterprise parameters for automated onboarding evaluation pipelines).

4.0 Open Integration & Support Center:

* 4.1 Open REST API Reference Terminal (Exposing endpoints like POST /api/v1/procurement/voice and webhook architectures).
* 4.2 Dynamic System Infrastructure Health Page (Streaming real-time latency graphs, endpoint status, and Langfuse tracing records).

5.0 Global Contact & Enterprise Lead Portal:

* 5.1 Dynamic Data Capture Form (Full Name, Corporate Email, Enterprise Classification Selector, Open Inquiry Box).
* 5.2 Regional Office Address Registry with responsive mapping frameworks.

2.2 Merchant Mobile Application (Flutter)

1.0 Enterprise Onboarding & Identity Authorization:

* 1.1 Branded Launch Splash Component.
* 1.2 Cellular Phone Identification Field paired with Hardware Token OTP Verification workflow nodes.
* 1.3 Store Registration Wizard (Capturing Store Name, Legal Commercial Registration ID, and sniped GPS device coordinates).
* 1.4 Hardware Permission Authorization Portal (Explicitly mapping iOS/Android OS Microphone access requests).

2.0 Core Workspace Hub (Persistent Bottom Navigation System):

* **2.1 Dashboard Overview (Home):**
  - 2.1.1 Omnipresent Floating Voice Capture Trigger (Massive orange/blue circular button centering the application layout).
  - 2.1.2 Low Stock Level Quick Warning Alert Card (#FFAB00 Dynamic Context Box).
  - 2.1.3 In-Flight Shipment Tracker Line (#36B37E Active Delivery Strip).
  - 2.1.4 Quick History Feed (Exposing the merchant's last 3 synchronized invoice transactions).
* **2.2 Local Inventory Matrix:**
  - 2.2.1 Product SKU Identification Matching Input Field.
  - 2.2.2 Live Asset Roster Grid (Displaying Item Title, Category Tag, Exact Current Quantities, and Unit Identifiers).
  - 2.2.3 Manual Stock Level Calibration Controller Interface.
* **2.3 Procurement Control Center:**
  - 2.3.1 In-Transit Shipment Tracking Pipeline Layout.
  - 2.3.2 Closed Historical Transaction Ledger Logs.
  - 2.3.3 Detailed Order Line-Item Workspace (Exposing item breakdowns, costs, and parent order routing associations).
* **2.4 Account Configuration & App Interface Matrix (Profile):**
  - 2.4.1 Legal Shop Profile Metadata & Identity Management.
  - 2.4.2 System Translation Control Toggle (Flipping UI layouts flawlessly between Arabic RTL and English LTR views).

3.0 Voice Procurement Processing Loop (The Core Path):

* 3.1 Floating Waveform Audio Overlay Container.
* 3.2 AI Microservice Processing Shimmer Loader State.
* 3.3 **Manual Correction UI / Order Review Hub:**
  - 3.3.1 Voice Note Transcription Reading Panes (Exposing raw colloquial Egyptian Arabic phrases).
  - 3.3.2 Editable Entity Form Fields Matrix (Item Name, Extracted Quantities Input, Urgency Level Boolean Switch).
  - 3.3.3 Dynamic Embedded Audio Player Widget (Sourced directly from VoiceProcurementLogs.RawAudioURL).
  - 3.3.4 Autonomous Agent Routing Distribution Summary Sheet (Exposing the calculated split distribution logic).
* 3.4 Order Settlement Confirmation Success Modal.

4.0 System Resilience Guard (Unhappy Path Fault Control UI):

* 4.1 Microphone Access Blocked Alternative Deep-Link Direct Guidance Route.
* 4.2 Local Offline Edge Gateway Synch Banner (#DE350B Alert detailing fallback to local Ollama nodes).
* 4.3 Low-Confidence AI Intent Extraction Prompt Framework (Manual Text Override Box Trigger).
* 4.4 Hard-Limit Procurement Violation Interface (Fraud Halt Alert Screen for anomalous volume or value bounds).

2.3 Supplier Web Dashboard (Angular - Enterprise Client)

1.0 Gateway Authentication Portal:

* 1.1 Supplier Portal Account Login View.
* 1.2 Automated Password Reset Verification Stream Handler.

2.0 Live Central Feed (Control Tower Dashboard Landing):

* 2.1 Real-Time Incoming Order Card Stack running on persistent WebSocket server loops.
* 2.2 Operational Financial Metric Panel Arrays (Aggregated Daily Billings, Total Pending Queue Loads, Active Stock Deficits).

3.0 Kanban Order Operations Board:

* 3.1 Pending Review Inbox Stack.
* 3.2 Accepted Processing Pipeline Stream.
* 3.3 Dispatched Logistics In-Transit Track.
* 3.4 Execution Exception Archive Column (Housing canceled, rejected, and fraud-halted entries).

4.0 RAG Document Knowledge Base Center:

* 4.1 Supplier Inventory Database Table Index (Base pricing control controls, SKU IDs, warehouse inventory indicators).
* 4.2 Drag-and-Drop Bulk File Ingestion Dropzone Portal (PDF/Excel Catalog Multi-Upload Stream).
* 4.3 Vector Chunk Inspector Frame (Exposing semantic overlap maps, metadata attributes, and pgvector upload verifications).

5.0 Fraud & Hard-Limits Configuration Dashboard:

* 5.1 Enterprise Risk Metrics Control Field Matrix (Direct entry paths for MaxOrderValue and MaxQuantityPerSKU validation parameters).
* 5.2 Active Flag Override Resolution Queue (List view detailing flagged order inputs awaiting administrative overrides).

6.0 Internal System Audit Room Dashboard:

* 6.1 Comprehensive Administrative Operation Record Matrix (Tracking ActionPerformed, TargetTable, and user ID hashes).
* 6.2 Identity Access Provisioning Hub (Enforcing Admin role scopes using strict least-privilege logic).
* 6.3 API Cryptographic Integration Key Vault Panel.

7.0 Business Intelligence Hub:

* 7.1 Financial Revenue Accrual Performance Visualization Components.
* 7.2 Product Category Volumetric Flow Analysis.

3. Granular Information Architecture (IA) Matrix

3.1 Public Web Structure Viewports

Viewport: Dynamic Infrastructure Status Page

* **Header Structural Element:** Standard public navbar containing structural directory paths, company marks, and localization dropdowns.
* **Real-Time Latency Analytics Graph Matrix:** Responsive line canvas layout tracing execution response metrics from the primary ITI API Access Gateway alongside secondary Groq API failover vectors.
* **Microservice Component Status Strip:** A vertical sequence of status blocks displaying real-time infrastructure states (Active, Degraded Operations, Offline Failover Enforced) leveraging system color codes (#36B37E, #FFAB00, #DE350B):
  - *System Block:* Voice Conversion Pipeline Module (Amazon Transcribe node health verification).
  - *System Block:* LLM Parsing Optimization Cluster (Claude Haiku/Sonnet gateway connection states).
  - *System Block:* Local Database Gateway Edge Architecture (Ollama synchronization capability state metrics).
* **Langfuse Tracing Dashboard Ledger:** Live operational grid tracing absolute token ingestion ratios, execution cost matrices, and background cyclical loop tracking times.

3.2 Mobile Interface Architecture Viewports

Viewport: AI Draft Correction / Review Portal (VoiceProcurementLogs Integration)

* **Header Section Structure:** Title string reading "Review Extracted Order Entities", backed by a prominent data field container rendering the NLPConfidenceScore metric inside a styled circular badge layout.
* **Embedded Raw Audio Playback Panel:** Native media widget bound directly to VoiceProcurementLogs.RawAudioURL. Includes integrated play/pause toggles and an interactive scrub bar enabling the merchant to review the exact audio recording interpreted by the backend.
* **Transcription Box Container:** Read-only block displaying the text string returned by Amazon Transcribe services (e.g., "[redacted] 20 [redacted] 5 [redacted]").
* **Editable JSON Entity Grid Matrix:** Dynamic form components mapping extracted entities directly into editable layout tables:
  - *Row Template Node:* Extracted Item Name Title (Text field input enabling manual structural corrections).
  - *Row Template Node:* Numeric Volume Input Field (Equipped with quick tap addition/subtraction step controls).
  - *Row Template Node:* Urgency Setting Switch Component (Boolean picker mapping priority tags directly to backend parameters).
* **Autonomous Split Routing Matrix:** Visual summary detailing the calculated split execution path generated by LangGraph agents:
  - *Supplier Split Card A:* Allocating Item A volumes to Supplier A based on lowest price rules.
  - *Supplier Split Card B:* Diverting Item B shortfalls to Supplier B to protect stock continuity metrics.
* **Primary Navigation Footers:** Primary Confirmation Action Node (#36B37E button executing SQL commitments), paired with a destructive Cancel Action Button (#DE350B).

Viewport: Order Detail View with Fragmented Split Routing Matrix

* **Header Section Container:** "Order Execution Profile", displaying the core Parent Order Identification String Hash along with a localized timestamp string.
* **Consolidated Order Invoice Summary:** Aggregated total pricing metrics, unique item counts, and current global transaction execution state chips (#FFAB00 for Drafts, #36B37E for Dispatched).
* **Fragmented Shipments Grid Roster:** An itemized series of cards exposing the exact operational architecture generated by OrderSplits data rows:
  - *Shipment Segment Block:* Unique Child Split ID string alongside the target Wholesaler Business Name field.
  - *Shipment Segment Block:* Dedicated line item breakdown grid displaying specific SKUs, assigned prices, and volume allocations within this specific shipment segment.
  - *Shipment Segment Block:* Individual Fulfillment State Chip Component tracking dispatch steps directly via the supplier (Pending Wholesale Review, Accepted/Assembled, In-Transit Logistics, Delivered Confirmation).

3.3 Web Interface Architecture Viewports

Viewport: Fraud Risk & Hard-Limits Setup Room (FraudPreventionLimits Integration)

* **Header Component Panel:** "Enterprise Procurement Guardroom - Rule Matrix Configuration Panel".
* **Risk Mitigation Parameters Entry Fields:** Structured form blocks with built-in validation rules that push updates directly to the database layer:
  - *Input Node:* Global Maximum Order Transaction Capital Cap (Validating monetary totals before order confirmation workflows can fire).
  - *Input Node:* Absolute Product Volume Cap Per SKU Unit (Determines the maximum quantity threshold allowed per item string inside a single order envelope).
  - *Input Node:* Dynamic Variance Exception Percentage Picker (Flags order spikes that depart from historical merchant ordering baselines).
* **System Action Rules Selector Array:** Dynamic select menus assigning programmatic actions when limits are tripped (Enforce Hard Block Shield, Route to Admin Verification Queue, Trigger Secondary Biometric Challenge).
* **Form Submission Row Control:** Primary execution action element mapping updates directly to the FraudPreventionLimits database configurations.

Viewport: Internal System Audit Room (SystemAuditLogs Integration)

* **Header Workspace Layout:** "Enterprise Platform Governance Ledger & Activity Trace".
* **Global Filter Workspace Strip:** Multi-choice dropdown blocks enabling administrators to slice system logging data by specific account identities, target database tables, or exact operation types (INSERT, UPDATE, RLS_BYPASS_ATTEMPT).
* **Master Forensic Audit Ledger Data Table:**
  - *Column Component 1:* System Audit Event Unique ID string hash.
  - *Column Component 2:* Actor Security Key Profile (Displaying full legal operator identity details and user role flags).

- *Column Component 3:* Action Execution Summary String (Exposing the exact operational logic executed against system assets).
- *Column Component 4:* Target Data Asset Reference (The specific database table name targeted by the interaction).
- *Column Component 5:* Trace Coordinates Mapping Node (Capturing client device system IP addresses and hardware signatures).
- *Column Component 6:* Chronological Execution Timestamp (Rendered explicitly in strict LTR English layout format).

4. Comprehensive Enterprise Application User Flows

Flow 1: End-to-End Autonomous AI Voice Ordering & Splitting Route

[Merchant Mobile] --> (Hold Mic Floating Action Button) --> (Stream Egyptian Colloquial Audio Note)

|

[SQL Database] <-- (Commit Transactional Records) <-- (Confirm UI) <-- (LangGraph Agent Split Result)

1. **Audio Ingestion Initiated:** The merchant opens the Flutter client, tapping and holding the large circular microphone action trigger. The UI responds by presenting a full-screen blurred interface overlay with an animated waveform visualization.
2. **Voice Note Pipeline Transport:** The merchant dictates their inventory requirements using natural, conversational Egyptian Arabic. Upon releasing the button, the raw audio stream binary is captured, packaged into an API payload, and pushed over to the .NET Core backend microservices layer.
3. **NLP Intent Processing:** The backend invokes the speech-to-text conversion layer via Amazon Transcribe services, routing the resulting text string through the institutional ITI API Access Gateway to be evaluated by Claude LLM engines. The system extracts entities with a high-accuracy baseline target of >92% for colloquial Arabic.
4. **Deterministic Optimization Agent:** The structured JSON payload is passed down to LangGraph orchestration engines. The agent checks the request against current wholesale price books, distributor catalog indexes, proximity variables, and real-time stock balances extracted from pgvector data.
5. **Agent Order Splitting Resolution:** If the algorithm discovers that the primary lowest-price distributor (Supplier A) cannot fulfill the total volume requested, the LangGraph agent splits the transaction. It allocates Supplier A's available stock and paths the remaining volume shortfall over to Supplier B's fulfillment pool.
6. **Correction UI Inspection Node:** The calculated split resolution is passed back up to the Flutter client layout framework, populating the Manual Correction UI workspace. The interface streams the media file container from VoiceProcurementLogs.RawAudioURL, prints the Arabic text transcription string, and renders the extracted items as editable form data rows.
7. **Transactional Core Finalization:** The merchant reviews the automated routing path and taps the primary "Confirm Order" action trigger. The mobile app fires a request to the .NET Core gateway API, which safely commits the transaction across the SQL Server database layers inside a protected transactional block.

Flow 2: Wholesaler Catalog RAG Vectorization Ingestion Cycle

[Supplier Web] --> (Dropzone PDF Multi-Upload Tool) --> (Transmit Document Binary File Streams)

|

[pgvector Engine] <-- (Cohere Rerank Semantic Chunk Index) <-- (Token Overlap Text Extraction)

1. **Document Binary Dispatch:** An administrator opens the Angular Wholesaler Control Tower portal, navigates over to the RAG Knowledge Base interface, and selects a bulk file inventory index dropzone component. They drop an unstructured inventory catalog asset (like a multi-page PDF document displaying 15,000 retail product stock variants) into the interface.
2. **Pipeline Ingestion Interceptor:** The web client reads the document file stream and transmits the payload over to the backend .NET Core upload endpoint through securely encrypted routes.
3. **Semantic Token Segmentation:** The processing microservice parses the text layouts using a semantic chunking methodology, establishing uniform 100-token text data overlaps to protect relationship signatures

surrounding specific SKU identifiers and unit prices.
4. **Vector Store Insertion Route:** The text data segments are transformed into dense multi-dimensional vector embeddings, routing the calculations directly into the pgvector database architecture.
5. **Retrieval Evaluation Cycle:** The system runs the indexes through Cohere Rerank 3.5 layers to verify data lookup accuracy and protect the AI matching algorithms from hallucinated outputs.
6. **Verification Workspace Summary:** The backend pushes an implementation completion signal back to the Angular client portal. The interface updates the processing table view, moving the target file status indicator chip from a yellow Processing loader into a green Vectorized via pgvector success badge.

Flow 3: Real-Time B2B Procurement Order Fulfillment Stream

[Supplier Dashboard] --> (Listen WebSocket Order Stream) --> (Toast Alert
Notification Triggered)

|

[Merchant Mobile] <-- (Fulfillment Push Notification) <-- (Deduct Stock) <--
(Supplier Clicks "Accept")

1. **Real-Time Data Pipeline:** A warehouse dispatcher opens the supplier dashboard client workspace, which connects to a persistent WebSocket server pipeline mapping data traffic directly from the .NET Core backend.
2. **Order Broadcast Event:** The moment a retail merchant taps the confirmation control node inside their mobile client app, the backend flags the specific shipment rows bound to this wholesaler business identity.
3. **UI Card Injection Event:** The supplier web client intercepts the transmission record over the open WebSocket connection, firing a crisp audio notification ping and adding a new transaction card into the incoming order stream column. The layout item displays the parent order ID, client details, order metrics, and an urgency alert badge.
4. **Fulfillment Action Choice:** The dispatcher reviews the line items and clicks the primary green "Accept" action button. The platform transmits an asynchronous command to the .NET Core backend, which validates warehouse availability rules and logs the inventory deductions.
5. **Cross-Platform Signal Broadcast:** The fulfillment framework fires a status update broadcast message out across the network. The merchant's mobile application intercepts the update, instantly transitioning the target shipment tracking chip from a yellow Pending Review status into a green In-Transit Logistics delivery layout card.

5. User Journey Blueprint Maps

Journey 1: Abd Elrahman (The SME Retail Merchant Perspective)

* **The Operational Friction:** Abd Elrahman is operating a high-volume local retail grocery outlet in a busy, noisy commercial street in Cairo. Multiple customers are crowded around the checkout counter, and he suddenly notices that his shelf stock for a primary cooking oil brand is completely empty. He has zero available time to log into a complex desktop web ERP system, handle dense search filter menus, or type manually into structured spreadsheet data tables.
* **The Remedial Action:** He opens the Salasel mobile application on his phone, taps and holds the prominent orange/blue voice procurement action trigger button, and speaks naturally: *"[redacted] 20 [redacted]"*.
* **The Internal AI Transition:** The mobile client captures the audio track, routes it to the backend for speech-to-text processing, extracts entities using Claude LLM engines, and splits the order parameters across distributors via LangGraph orchestrations. The processed results update on his phone screen in under two seconds.
* **The Resolution Moment:** He reviews the clear, organized layout in the Manual Correction UI, plays the audio note clip container to verify his request, checks the itemized entries, and taps the primary confirmation button. The system commits the transaction to the database, allowing Abd Elrahman to focus entirely on his customers without navigating frustrating technical frameworks.

Journey 2: Delta Wholesale (The Regional Supplier Admin Perspective)

* **The Operational Friction:** The warehouse floor operators at Delta Wholesale are constantly struggling to manage a chaotic flood of unstructured WhatsApp text chains, handwritten order pads, and phone logs from retail merchants. This disconnected workflow leads to inventory logging mistakes, delayed shipping schedules, and lost revenue.
* **The Remedial Action:** The enterprise transitions its order tracking workflows onto the Salasel platform and leaves the central dashboard portal open on a terminal screen on the main warehouse floor.
* **The Internal AI Transition:** Instead of manually translating unverified messages, the system streams verified order entries directly into their incoming Kanban view via stable real-time WebSocket connections.
* **The Resolution Moment:** An audible notification sound plays as Abd Elrahman's split order data drops into the inbound dashboard queue. The fulfillment supervisor reviews the clean interface data card and clicks the primary green "Accept" action link. The system automatically prints a shipping manifest, adjusts warehouse inventory balances, and sends a push notification back to the merchant's mobile device.

6. Complete Edge-Case, Exception & Error State Roster

6.1 System Exception: Device Hardware Permission Blocked UI

* **The Trigger Event:** The merchant attempts to initiate a voice order sequence, but the mobile OS intercepts the command and returns an error because microphone capture access permissions were blocked or denied during application setup.
* **The Interface Resolution:** The client application closes the standard audio recording panel and surfaces a clean, full-screen hardware exception alert state. The layout explains exactly why microphone access is required for conversational procurement workflows. It features a single primary action link button that deep-links directly into the mobile device's native OS application settings panel, clearing a frictionless recovery path for the user.

6.2 System Exception: Network Outage & Offline Edge Gateway Fallback

* **The Trigger Event:** The merchant is in the middle of dictating an active voice procurement message when the local cellular network drops offline due to a data routing failure.
* **The Interface Resolution:** The Flutter application displays a persistent red system warning banner at the top of the viewport structure reading "No Internet Connection - Local Edge Processing Active". Rather than breaking the workflow, the app redirects the audio payload over the local network to an on-site edge gateway router device. A self-hosted local Ollama Llama 3 model processes the transcription and entity extraction, caching the results inside an offline transaction queue until cloud synchronization services are restored.

6.3 System Exception: Low-Confidence AI Speech Ingestion Exception

* **The Trigger Event:** The merchant records a voice note from a noisy retail floor, causing loud background sound artifacts that drop the transcription accuracy metric below the platform's strict 92% Arabic NLP baseline confidence score.
* **The Interface Resolution:** The client UI opens an explicit diagnostic resolution modal container. It shows a warning note reading "Audio input unclear due to ambient noise. Please review the text transcription, retry recording, or switch to manual typing." The modal presents a text input component pre-populated with the low-confidence string translation, enabling the merchant to quickly clean up the entries manually without losing their order draft.

6.4 System Exception: Enterprise Procurement Fraud Detection Trip

* **The Trigger Event:** A merchant account submits a voice note request containing an outlier quantity value that violates established ordering baselines (e.g., requesting 10,000 cases of an item instead of 10), tripping the backend system's hard limits.
* **The Interface Resolution:** The Manual Correction UI halts standard processing, replaces the submission button with a locked warning state, and tints the layout view with a critical red error hue (#DE350B). The screen flashes an alert message reading "Order transaction volume flags hard system constraints. Sent to administrative overview for approval." The transaction state parameter is updated to Fraud Flagged across the database, routing the line items to a supervisor review queue in the administration portal.